

Yucheng (Cecilia) Zhang

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EDUCATION

Northwestern University

M.S. in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

B.S. in Mathematics, Minor in Computer Science / GPA: 3.74/4.00

Aug. 2023 – May 2026

Austin, TX

TECHNICAL SKILLS

Machine Learning: Scikit-learn, PyTorch, TensorFlow, classification, feature engineering, model evaluation, ROC/AUC, backtesting

Generative AI: OpenAI, Gemini, Claude, LangChain, CrewAI, prompt design, context filtering, LLM evaluation

Data Science: Pandas, NumPy, R, MATLAB, MNE, ggplot2, statistics, permutation-based FDR, visualization

Data/Cloud: SQL Server, PostgreSQL, Azure Synapse, Databricks, Delta Lake, AWS, Docker, Power BI, Linux

Domains: Credit risk, fraud detection, bioinformatics, EEG/fNIRS, computer vision, financial analysis

MACHINE LEARNING, AI, AND DATA EXPERIENCE

CodeAssist – AI Feedback System | GitHub

Aug. 2025 – Present

University of Texas at Austin / Flask, React, PostgreSQL, OpenAI, Gemini, Claude

Austin, TX

- Built assignment-level AI feedback controls for prompt templates, context permissions, model settings, request limits, and cooldown policies.
- Implemented LLM context-filtering logic so only instructor-approved assignment descriptions, student code, test results, test cases, and student output are sent to AI providers.
- Standardized multi-provider AI workflows across OpenAI, Gemini, Claude, and local models, including model selection, API configuration, and timeout handling.

Enterprise Risk Analyst Intern

May 2025 – Dec. 2025

Toyota Industries Commercial Finance / Python, SQL, Azure, Power BI

Dallas, TX

- Built an end-to-end credit-risk modeling pipeline from SQL Server data, spanning ETL, credit-feature engineering, Logistic Regression, Random Forest training, and validation.
- Engineered features including PayNet inquiries, loan-to-value ratios, account tenure, and asset-segment signals for trucks, trailers, and used-vehicle loans.
- Validated models with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** and improving high-risk case identification by approximately **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups and informing credit scorecard refinements.

Researcher & TA – Biological Data Analytics

Jan. 2024 – Dec. 2024

University of Texas at Austin / R, Linux, AWS, Docker

Austin, TX

- Processed **250GB** of RNA-seq data with Salmon, RSEM, Trinity, and samtools, achieving a **98.7% mapping rate** across reproducible pipelines.
- Performed Voom/Limma differential-expression analysis, identifying 48 key genes across time-series samples with permutation-based FDR.
- Automated normalization, PCA, and volcano-plot workflows in R and ggplot2, improving interpretation efficiency by 15%.

FinChat – AI-Powered Financial Analysis System

Jan. 2023 – May 2024

Southern Methodist University / Python, FinGPT, LangChain, CrewAI

Remote

- Built LLM workflows for financial report analysis, extracting and processing 100+ SEC filings from EDGAR into a scalable data lake.
- Integrated FinGPT, LangChain, and CrewAI components for metric extraction, trend detection, and contextual financial interpretation, improving filing-analysis consistency by 3%.

Data Scientist Assistant Intern – Image Restoration

Jul. 2023 – Sep. 2023

Chinese Academy of Sciences / TensorFlow, Cloud, GFPGAN

Beijing, China

- Deployed image restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Constructed a TensorFlow 2 CNN, split 3.4k images into train/validation/test sets, and achieved **80% accuracy** across 5 image categories.

RESEARCH PROJECTS AND PUBLICATIONS

Neuroengineering Lab – Machine Learning Research Assistant

Jan. 2023 – May 2024

Python, MATLAB, MNE, RF, SVM, KNN, XGBoost

Remote

- Processed **35GB** of EEG/fNIRS data for signal cleaning, feature extraction, connectivity analysis, and statistical testing; contributed to Frontiers and IEEE publications.
- Evaluated emotion-recognition models using RF, SVM, KNN, and XGBoost to compare model behavior across EEG-derived features and experimental conditions.

Kiwifruit Quality Grading – Machine Learning Member | Paper

Aug. 2022 – May 2024

Computer Vision, Feature Fusion, Model Evaluation

Published 2025

- Contributed to image-based grading with color, texture, shape, and fusion features; benchmarked models for a paper published in *Expert Systems with Applications*.
- Compared feature combinations to optimize grading accuracy and interpret model behavior for the paper's ML methodology section.

SELECTED PUBLICATIONS

- Expert Systems with Applications* (2025): kiwifruit quality grading using image multi-feature fusion and ResTNet.
- Frontiers in Neurology* (2024) and *Frontiers in Neuroscience* (2023): EEG/fNIRS cognitive impairment and digital therapy studies.